

✓ IT Helpdesk Ticket Analytics

Python | Pandas | Matplotlib | Seaborn

Project Overview

This project analyzes an IT helpdesk support ticket dataset to identify ticket trends, department workload, and ticket priority distribution. The analysis includes data cleaning, exploratory data analysis (EDA), and visualization using Python.

✓ Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- SQLite
- Google Sheets

```
!pip install wordcloud
```

```
Requirement already satisfied: wordcloud in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: numpy>=1.19.3 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: pillow in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-pack
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dis
Requirement already satisfied: cyclor>=0.10 in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/di
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/di
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dis
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packag
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from wordcloud import WordCloud
import sqlite3
from collections import Counter
```

```
plt.style.use("ggplot")
```

Load Dataset

```
from google.colab import drive
drive.mount('/content/drive')
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount(

Dataset Overview

```
df = pd.read_csv('/content/drive/MyDrive/IT Helpdesk Analytics/ticket.csv')
display(ticket_df.head())
```

Unnamed: 0		Body	Department	Priority	Tags
0	0	Dear Customer Support Team,I am writing to rep...	Technical Support	high	['Account', 'Disruption', 'Outage', 'IT', 'Tec...
1	1	Dear Customer Support Team,I hope this message...	Returns and Exchanges	medium	['Product', 'Feature', 'Tech Support']
		Dear Customer			['Billing', 'Payment']

```
df.shape
```

(29651, 5)

```
df.columns
```

```
Index(['Unnamed: 0', 'Body', 'Department', 'Priority', 'Tags'], dtype='object')
```

```
df.rename(columns={
    "Unnamed: 0": "Ticket_ID"
}, inplace=True)
```

```
df.head()
```

	Ticket_ID	Body	Department	Priority	Tags
0	0	Dear Customer Support Team,I am writing to rep...	Technical Support	high	['Account', 'Disruption', 'Outage', 'IT', 'Tec...
1	1	Dear Customer Support Team,I hope this message...	Returns and Exchanges	medium	['Product', 'Feature', 'Tech Support']
2	2	Dear Customer Support Team I hope	Billing and	low	['Billing', 'Payment', 'Account']

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 29651 entries, 0 to 29650
Data columns (total 5 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Ticket_ID   29651 non-null  int64
1   Body        29650 non-null  object
2   Department  29651 non-null  object
3   Priority     29651 non-null  object
4   Tags        29651 non-null  object
dtypes: int64(1), object(4)
memory usage: 1.1+ MB
```

```
df.describe(include="all")
```

	Ticket_ID	Body	Department	Priority	Tags
count	29651.000000	29650	29651	29651	29651
unique	NaN	25055	10	3	12946
top	NaN	The data analytics tool experiences sluggish p...	Technical Support	medium	['Bug', 'Performance', 'IT', 'Tech Support']
freq	NaN	2	8617	12126	878
mean	14825.000000	NaN	NaN	NaN	NaN
std	8559.650752	NaN	NaN	NaN	NaN
min	0.000000	NaN	NaN	NaN	NaN
25%	7412.500000	NaN	NaN	NaN	NaN
50%	14825.000000	NaN	NaN	NaN	NaN
75%	22237.500000	NaN	NaN	NaN	NaN

▼ Data Cleaning

```
df.isnull().sum()
```

	0
Ticket_ID	0
Body	1
Department	0
Priority	0
Tags	0

```
dtype: int64
```

```
df.fillna("Unknown",inplace=True)
```

```
df.isnull().sum()
```

	0
Ticket_ID	0
Body	0
Department	0
Priority	0
Tags	0

dtype: int64

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df.drop_duplicates(inplace=True)
```

```
df["Ticket_ID"].duplicated().sum()
```

```
np.int64(0)
```

```
df["Department"].unique()
```

```
array(['Technical Support', 'Returns and Exchanges',
      'Billing and Payments', 'Sales and Pre-Sales',
      'Service Outages and Maintenance', 'Product Support', 'IT Support',
      'Customer Service', 'Human Resources', 'General Inquiry'],
      dtype=object)
```

```
len(df)
```

```
29651
```

```
df["Department"].nunique()
```

```
10
```

```
df["Priority"].unique()
```

```
array(['high', 'medium', 'low'], dtype=object)
```

✓ Exploratory Data Analysis (EDA)

Analysis 1: Ticket Distribution by Department

```
department_count=df["Department"].value_counts()
```

```
department_count
```

	count
Department	
Technical Support	8617
Product Support	5539
Customer Service	4482
IT Support	3500
Billing and Payments	3017
Returns and Exchanges	1467
Service Outages and Maintenance	1157
Sales and Pre-Sales	885
Human Resources	568
General Inquiry	419

```
dtype: int64
```

```
plt.figure(figsize=(10,6))

sns.countplot(
    data=df,
    y="Department",
    order=df["Department"].value_counts().index,
    palette="viridis"
)

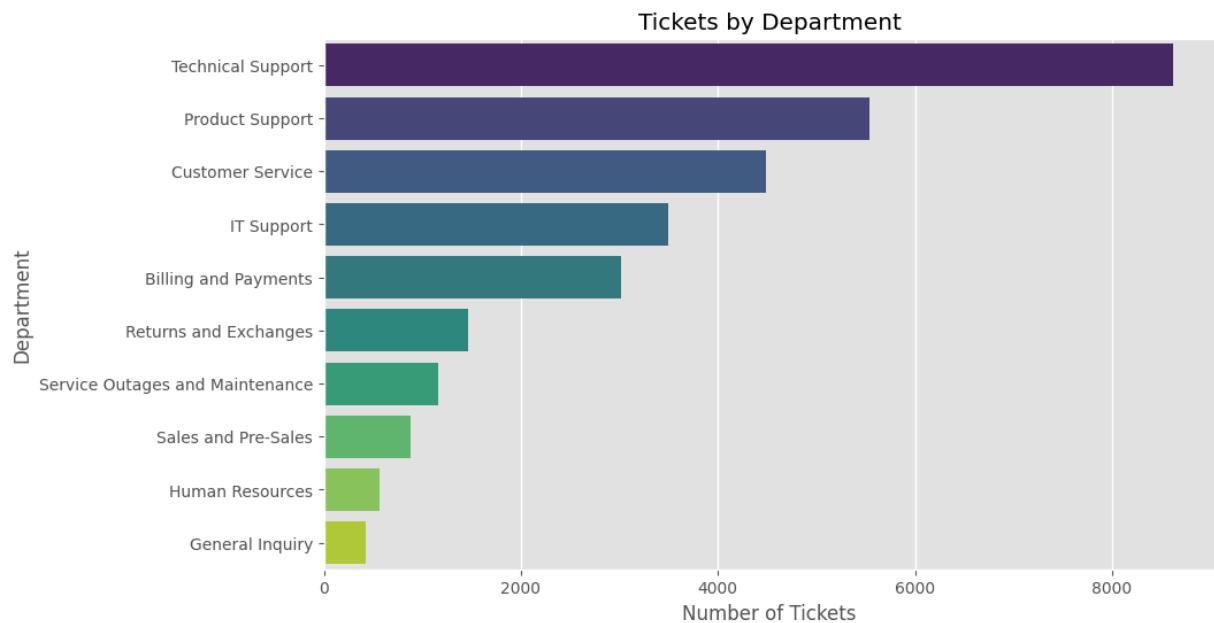
plt.title("Tickets by Department")
plt.xlabel("Number of Tickets")
plt.ylabel("Department")

plt.show()
```

```
/tmp/ipykernel_777/3635485703.py:3: FutureWarning:
```

```
Passing `palette` without assigning `hue` is deprecated and will be removed in v
```

```
sns.countplot(
```



Business Insight

Technical Support received the highest number of support tickets (8,617), indicating the greatest operational workload.

Recommendation

Increase staffing or automate repetitive requests to improve response time.

✓ Analysis 2: Ticket Priority Distribution

```
priority_count=df["Priority"].value_counts()
```

```
priority_count
```

	count
Priority	
medium	12126
high	11512
low	6013

dtype: int64

```
plt.figure(figsize=(7,5))

sns.countplot(
    data=df,
    x="Priority",
    palette="Set2"
)

plt.title("Priority Distribution")

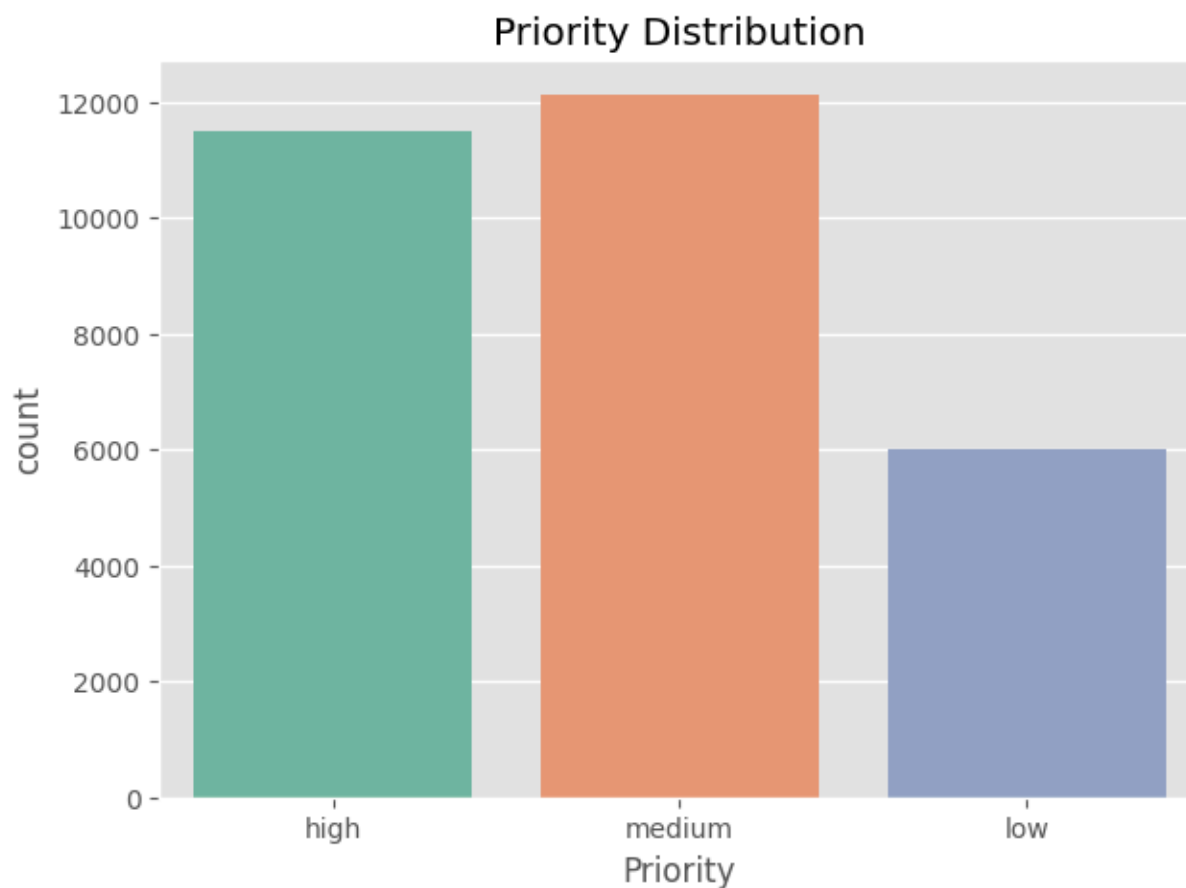
plt.show()
```



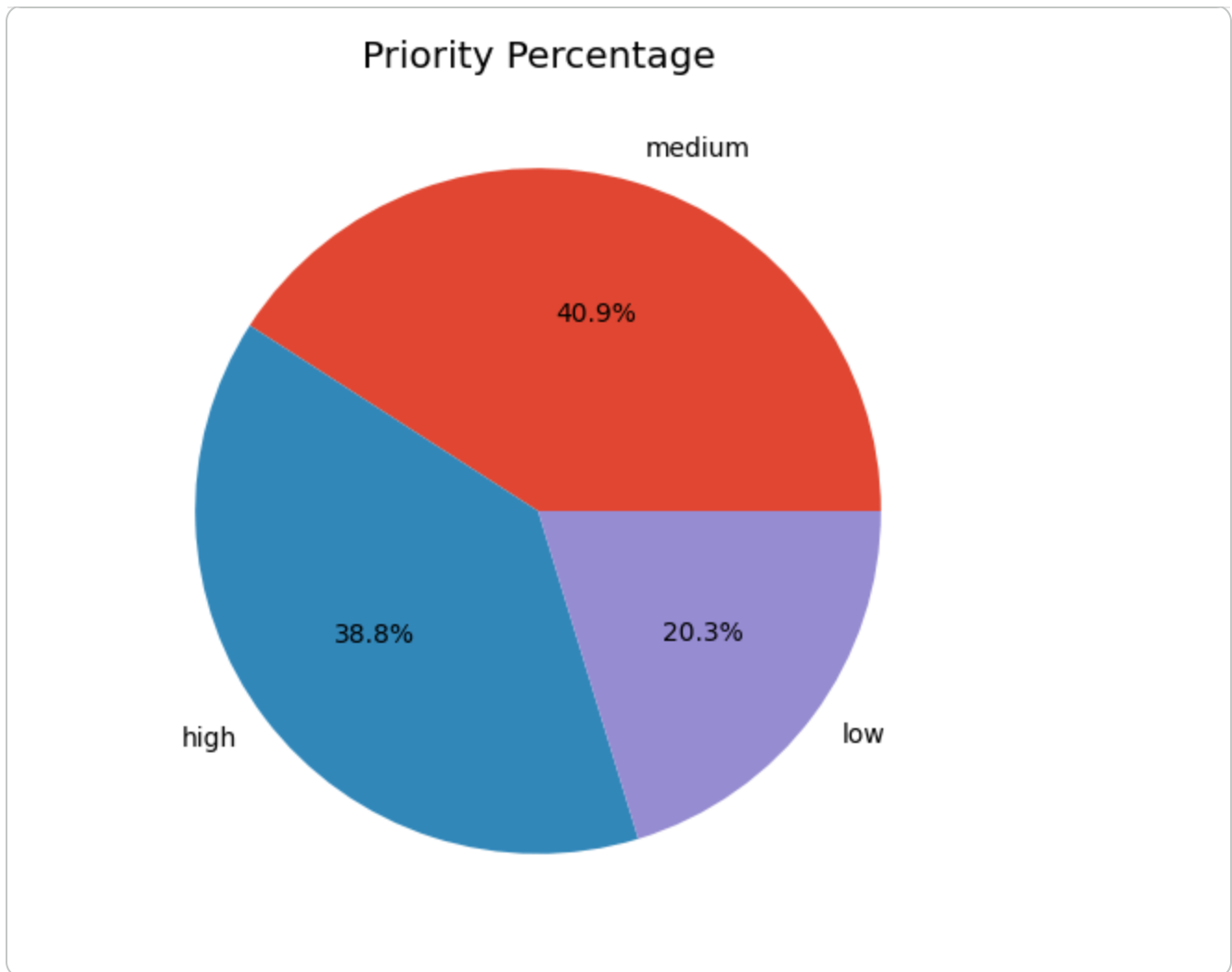
```
/tmp/ipykernel_777/2429828299.py:3: FutureWarning:
```

```
Passing `palette` without assigning `hue` is deprecated and will be removed in v
```

```
sns.countplot(
```



```
priority_count.plot(  
    kind="pie",  
    autopct="%1.1f%%",  
    figsize=(6,6)  
)  
  
plt.ylabel("")  
  
plt.title("Priority Percentage")  
  
plt.show()
```



Business Insight

Medium-priority tickets represent the largest share of all requests.

Recommendation

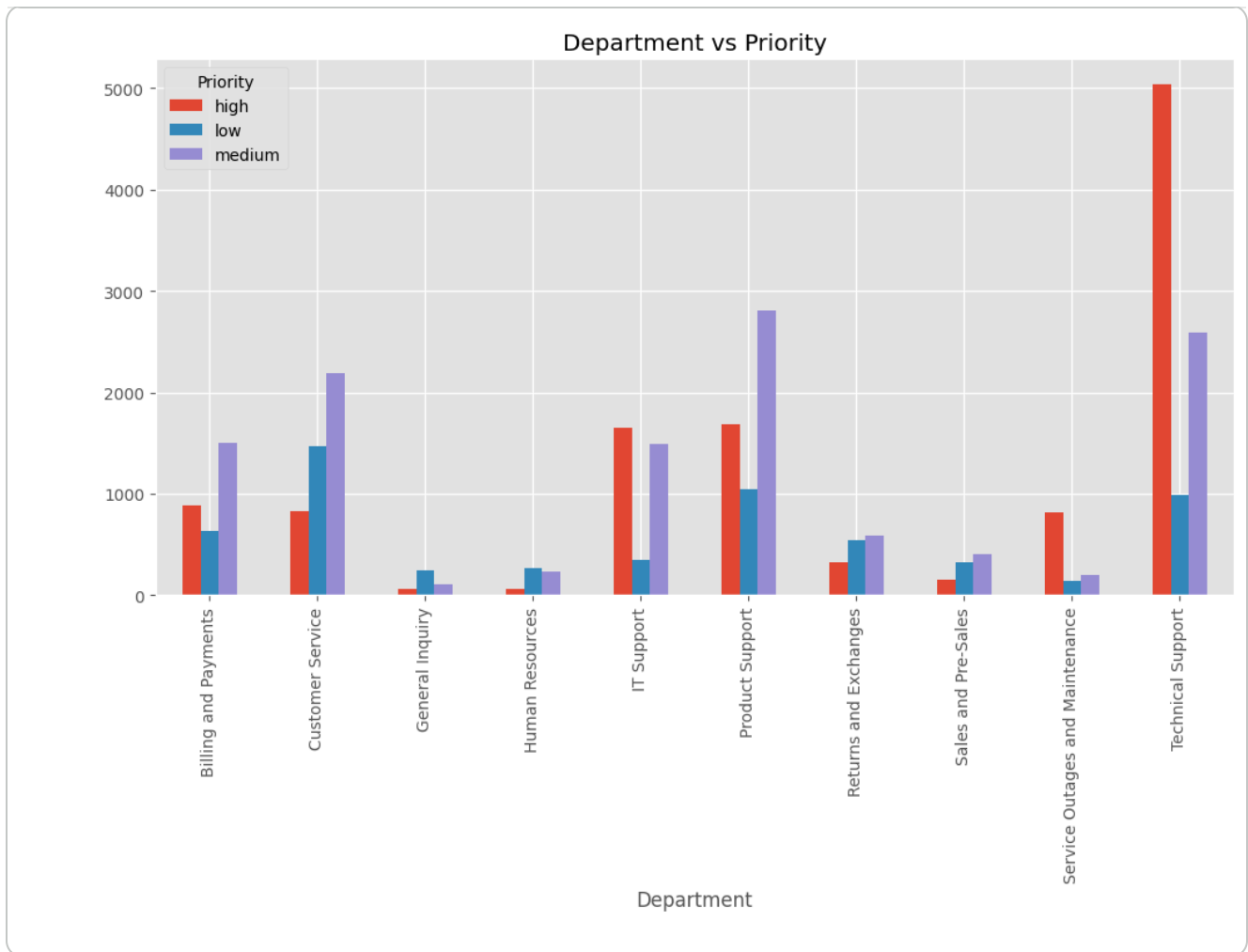
Automating medium-priority requests can reduce workload and improve response time for high-priority tickets.

✓ Analysis 3: Department-wise Priority Analysis

```
cross=pd.crosstab(  
    df["Department"],  
    df["Priority"]  
)  
  
cross
```

Department	Priority		
	high	low	medium
Billing and Payments	886	632	1499
Customer Service	828	1464	2190
General Inquiry	66	246	107
Human Resources	59	271	238
IT Support	1657	347	1496
Product Support	1682	1048	2809
Returns and Exchanges	328	546	593
Sales and Pre-Sales	154	323	408
Service Outages and Maintenance	818	143	196
Technical Support	5034	993	2590

```
cross.plot(  
    kind="bar",  
    figsize=(12,6)  
)  
  
plt.title("Department vs Priority")  
  
plt.show()
```



Business Insight

Technical Support handles the highest number of high-priority tickets, indicating that it manages the most critical issues.

Recommendation

Allocate additional engineers to the Technical Support department.

✓ Analysis 4: Department-wise Ticket Percentage

```
department_percent = (
    df["Department"]
    .value_counts(normalize=True)
    *100
)

department_percent
```

proportion	
Department	
Technical Support	29.061414
Product Support	18.680652
Customer Service	15.115848
IT Support	11.803986
Billing and Payments	10.175036
Returns and Exchanges	4.947557
Service Outages and Maintenance	3.902061
Sales and Pre-Sales	2.984722
Human Resources	1.915618
General Inquiry	1.413106

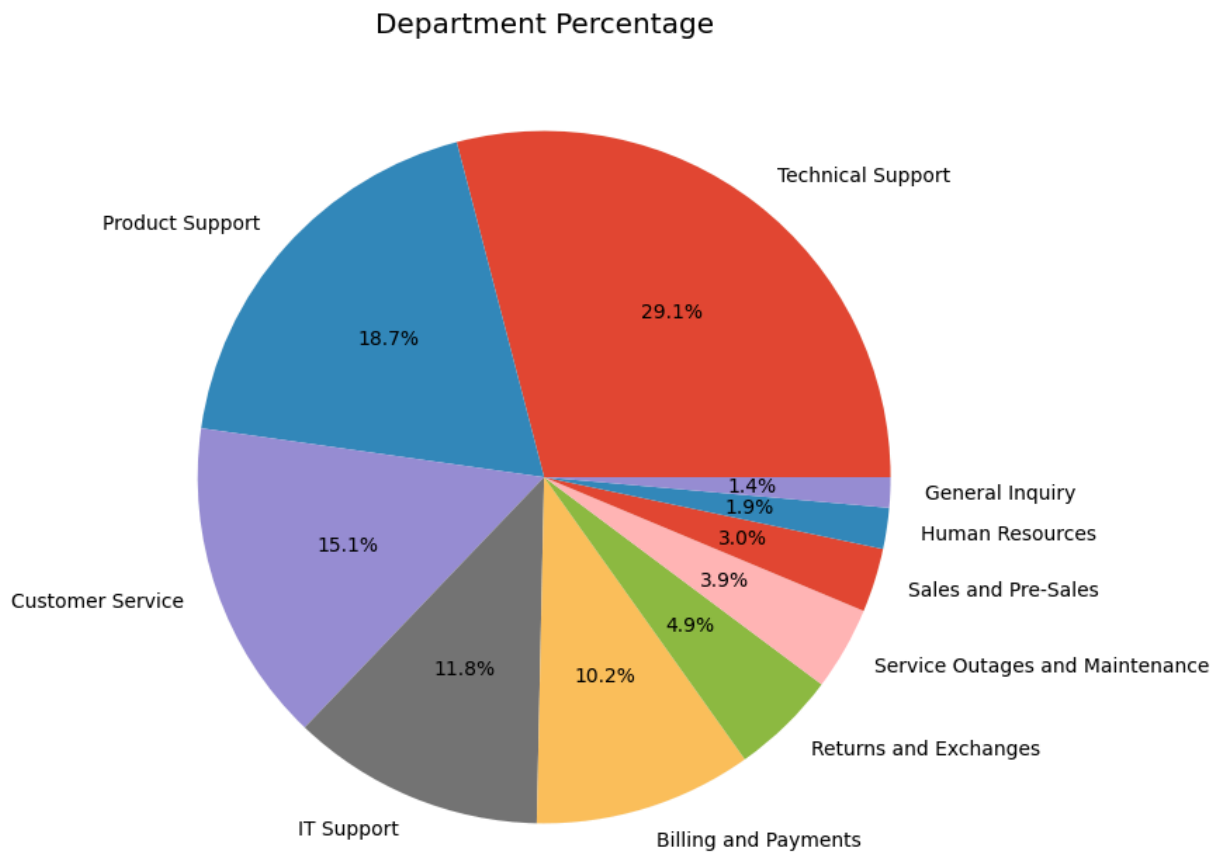
dtype: float64

```
department_percent.plot(
    kind="pie",
    autopct="%1.1f%%",
    figsize=(8,8)
)

plt.ylabel("")

plt.title("Department Percentage")

plt.show()
```



✓ Business Insight

Technical Support contributes the largest percentage of total tickets, followed by Product Support and Customer Service.

This highlights where the majority of operational workload is concentrated.

```
priority_percent = (  
    df["Priority"]  
    .value_counts(normalize=True)  
    *100  
)
```

priority_percent

proportion	
Priority	
medium	40.895754
high	38.824997
low	20.279249

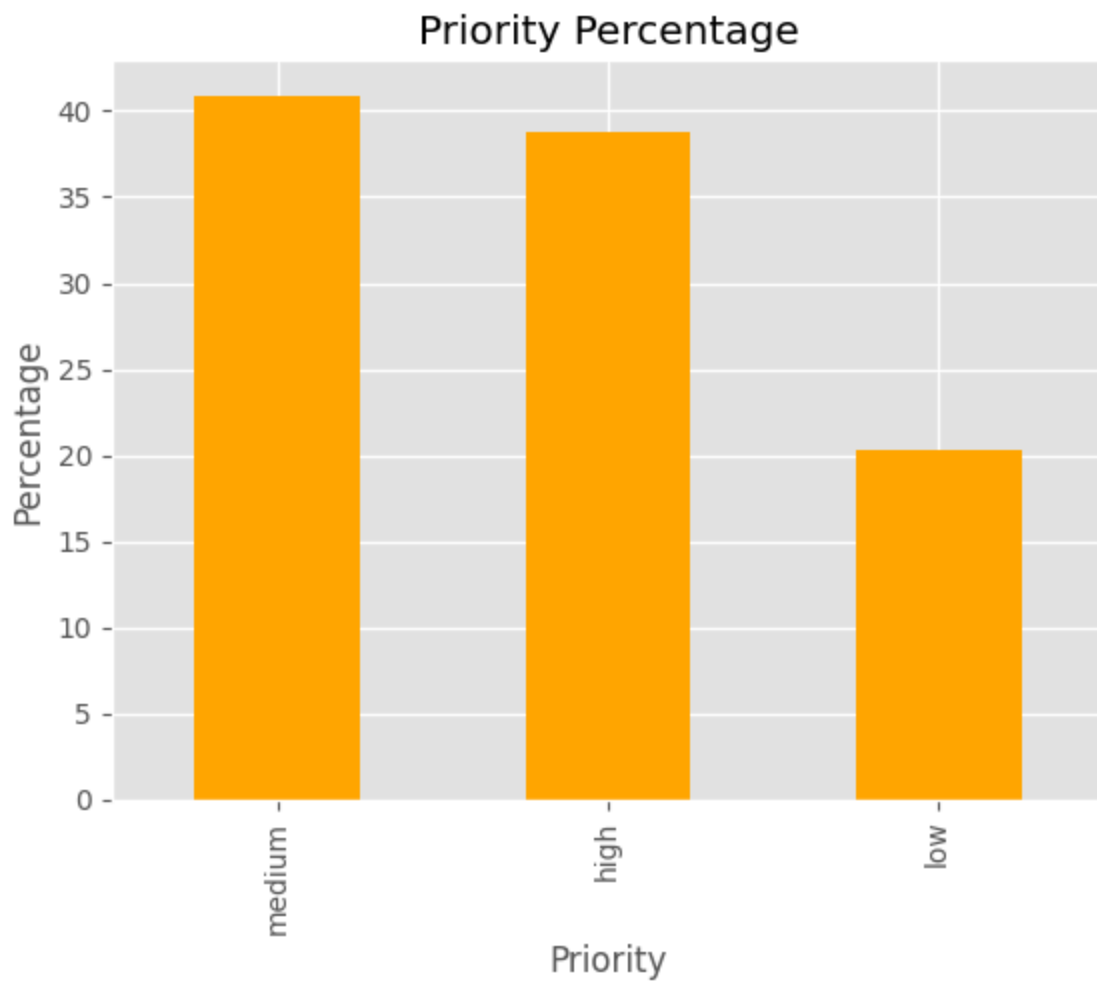
dtype: float64

```
priority_percent.plot(
    kind="bar",
    color="orange"
)

plt.title("Priority Percentage")

plt.ylabel("Percentage")

plt.show()
```



```
tag_count=tags["Tags"].value_counts()
```

```
tag_count.head(20)
```


count	
Tags	
Tech Support	14152
IT	13993
Performance	11410
Feedback	8716
Documentation	7644
Bug	7532
Security	6596
Feature	5777
Disruption	4244
Outage	3593
Technical	3530
Network	3250
Product	2755
Sales	2605
Resolution	2414
Guidance	1973
Recovery	1764
Billing	1738
Crash	1725
Integration	1382

dtype: int64

